Sahaj Bhadja

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Summary

Software engineer with 2+ years of experience shipping **AI-powered full-stack products** across **Python, React, Next.js, Tailwind, and SQL**, building responsive user experiences, background-job workflows, and production backends for high-usage applications.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies

PDEU
B.Tech. in Computer Science and Engineering *Nov 2020 – May 2024*
GPA: 3.88/4 Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Built a multi-tenant AI product across **FastAPI**, **Azure SQL**, and **PostgreSQL**, supporting 500+ users while reducing dashboard latency from 1.2s to 350ms through schema, indexing, and query optimization.
- Implemented fault-tolerant background-job and integration flows for 5K+ monthly jobs with validation, reconciliation checks, and release safeguards, improving throughput 3x under concurrent load.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Developed a user-facing AI grading platform with **Flask** + **React**, secure file ingestion, and modular APIs, handling 10K+ daily grading requests for 150+ instructors.
- Built concurrent processing and rubric-generation workflows on **AWS RDS** with CI/CD-backed releases, cutting feedback turnaround time by 60% while maintaining stable production performance.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python-based experimentation and simulation pipelines for agent-style autonomous-planning models, processing 100K+ trajectory logs and improving long-horizon planning performance by 18%.
- Collaborated on model behavior analysis, debugging, and evaluation workflows, improving closed-loop stability by 15% while documenting results for iterative research and product-style experimentation.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered microservice-based APIs and cross-platform data workflows with **Python**, Jenkins, and AWS services, improving response times by 30% and pipeline efficiency by 68% in production environments.
- Maintained 20+ unit and integration suites with PyTest and Mockito across Windows, Linux, and macOS, strengthening release reliability and code quality in Agile delivery cycles.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN-based operations platform with **REST APIs**, WebSockets, and microservices, supporting 8+ business workflows and 2M+ synchronized records in a user-facing system.
- Implemented CI/CD with Jenkins and Docker plus modular auth, billing, and notification services, reducing deployment time by 50% and sustaining 95%+ successful build validation.

Skills

Languages: Python, Java, C++, JavaScript/TypeScript, SQL
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, JSON, OOP, Background Jobs, Auth Flows
Frontend: React.js, Next.js, TailwindCSS, HTML/CSS, Streaming UI Patterns, State Management
Databases: PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL, Relational Data Modeling, Caching Concepts
Cloud & Infrastructure: AWS (RDS, S3, Glue), Microsoft Azure, Docker, Kubernetes, Jenkins, Git/GitHub
AI/ML: LLM Systems, Prompt Engineering, Transformers, AI Coding Tools, Code Generation, Agentic Workflows
Engineering Practices: CI/CD, Unit Testing, Integration Testing, Performance Profiling, Agile, System Design

Projects

PiXMod | *Next.js, TailwindCSS, MongoDB, Stripe, Transformers*
- Built an AI-powered SaaS product with a responsive **Next.js** interface, image-processing workflows, and subscription billing, increasing premium conversion by 18% while supporting scalable storage and retrieval in MongoDB.

Chat2SQL | *Java, TypeScript, SQL, Docker, Maven*
- Built an AI-assisted SQL workspace across a **Java** + **TypeScript** codebase, supporting 10+ databases and accelerating query authoring, schema exploration, and structured execution in a developer-focused product.

Big-Brain | *LLaMA-2, FAISS, PostgreSQL, GCP*
- Built an AI document product with semantic retrieval, summarization, and real-time collaboration across 50K+ documents, improving retrieval latency by 40% through optimized indexing and data-ingestion pipelines.